



A hybrid machine learning framework for analyzing human decision-making through learning preferences^{☆,☆☆}

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ABSTRACT

Multiple criteria decision aiding (MCDA) is a family of analytic approaches to depicting the rationale of human decisions. To better interpret the contributions of individual attributes to the decision maker, the conventional MCDA approaches assume that the attributes are monotonic and preference independence. However, the capacity in describing the decision maker's preferences is sacrificed as a result of model simplification. To meet the decision maker's demand for more accurate and interpretable decision models, we propose a novel hybrid method, namely **Neural Network-based Multiple Criteria Decision Aiding (NN-MCDA)**, which combines MCDA model and machine learning to achieve better prediction performance while capturing the relationships between individual attributes and the prediction. NN-MCDA uses a linear component (in an additive form of a set of polynomial functions) to characterize such relationships through providing explicit non-monotonic marginal value functions, and a nonlinear component (in a standard multilayer perceptron form) to capture the implicit high-order interactions among attributes and their complex nonlinear transformations. We demonstrate the effectiveness of NN-MCDA with extensive simulation studies and three real-world datasets. The study sheds light on how to improve the prediction performance of MCDA models using machine learning techniques, and how to enhance the interpretability of machine learning models using MCDA approaches.

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1. Introduction

Multiple criteria decision aiding (MCDA) has been a fast-growing area of operational research during the last several decades [1]. It involves a finite set of alternatives (e.g. actions, items, policies) that are evaluated from a set of conflicting multiple criteria or attributes¹. The Decision Maker's (DM's) decision is assumed to be driven by his/her underlying *global value (utility) function* [3]. This global value measures the DM's desirability for an alternative and can be disaggregated into a set of per-attribute *marginal value functions* that represent the DM's evalua-

tion of the corresponding attribute. These marginal value functions can be learned by the DM's judgments on learning examples (e.g. pairwise comparison between two alternatives). Once the marginal value functions are deciphered, we can understand the decision-making rationale, based on which we can predict the judgment of the DM. This process is referred to as the preference disaggregation approaches of MCDA.

The value function-based preference disaggregation approaches of MCDA have been successfully applied to many scenarios, such as consumer preference analysis [4], financial decisions [5], nanoparticles synthesis assessment [6], and medical therapy [7]. They can provide a comprehensive preference model that uses a value function to interpret the detailed relationship between each attribute and outcome. In this study, the interpretability refers to the model capacity to provide such explicit value functions. This explicit form of contributions helps the DM exploit and correct the obtained patterns and rules with prior knowledge, facilitate the downstream managerial decision-making, and incorporate critical domain knowledge [8].

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¹ In machine learning, criteria refer to attributes or features with preference order scales [2]. For consistency, we use "attribute" in this paper.

To enhance model interpretability, the conventional preference disaggregation MCDA approaches assume that the attributes are monotonic and preference independence [3]. The first assumption ensures the value functions to be in a predefined simple shape and the other assumption allows the preference model to be additive. These assumptions avoid obtaining an extremely complex preference model, making it easier to be interpreted by the DM. However, the capability of the conventional preference disaggregation approaches in describing the DM's preference is sacrificed as a result of the model simplification. Moreover, for many management problems, even if the attributes are well-chosen, the models obtained under these assumptions are often not fully applicable because these attributes are inherently non-monotonic or they interact with other attributes. The conventional approaches can relax the constraints/axioms, however, it comes at the cost of obtaining a model that is less explicit for interpretation [9–11].

Recently, many novel models have been proposed to generalize the value function-based preference disaggregation approaches of MCDA. Considering interacting attributes, [11] and [12] account for positive and negative interactions among attributes, and add interaction terms to the global value function for each alternative. They require the DM to provide some knowledge about the interacting pairs that are mined by the models. These studies only consider the interaction between pairs of attributes because higher-order interactions require more cognitive efforts and more computational cost. To describe preference inflexions, Ghaderi et al. [10] and Liu et al. [9] propose to constrain on variations of the slope to obtain non-monotonic marginal value functions without serious overfitting problem. Both of their approaches obtain non-smooth value functions in which it is difficult to interpret attitudes towards risks due to the use of non-derivative functions. Since a differentiable marginal value function is essential to analyze consumer behavior, [13] utilize semidefinite programming to infer the key parameters for monotonic polynomial marginal value functions. Nowadays, MCDA approaches are expected to deal with a large amount of data in many disciplines. Liu et al. [9] embed the MCDA approach into a regularization framework to approximate marginal value functions in any piece-wise linear shapes, and provide efficient algorithms to handle larger learning sets. Comparing with these recent advances, this study aims to solve the aforementioned challenges by providing a machine learning-enabled non-monotonic and more expressive MCDA approach for complex real-world applications.

Machine learning has recently been widely adopted to address challenging decision-making problems in a variety of managerial contexts such as marketing [14], credit-risk evaluation [15] and healthcare management [16]. Many popular machine learning models such as neural networks are *black-box*, and thus are not readily interpretable to the DM. Recent researches have proposed approaches to addressing the interpretability of machine learning models [17], such as the gradient-based methods [18,19], the sensitivity analysis [20,21], and the mimic models [15,16]. Among these methods, a particular important stream of research focuses on learning explicit functions to characterize the nonlinear contributions of variables during the training process. A typical example is the generalized additive model (GAM), which uses a link function to build a connection between the mean of the prediction and a function of the variables [22]. It is good at both dealing with and presenting the nonlinear and non-monotonic relationship between the variables and the prediction [23]. Therefore, GAM is usually more accurate than linear additive models but difficult to be interpreted if the underlying function shapes are extremely complicated, and thus cannot be easily understood by the DM.

MCDA and machine learning can provide natural benefits to each other [8]. On one hand, machine learning techniques are capable of handling high-dimensional and nonlinear data because of the high complexity of the model. They can help MCDA approaches relax conventional assumptions and improve model performance.

On the other hand, the preference disaggregation approaches of MCDA use a global value function (usually in an additive form with a predefined shape) to reveal the rationale of DM's judgment. Thus, MCDA approaches provide convincing evidence to assist comprehending the decision-making. They can help enhance the interpretability of "black-box" machine learning models.

This paper proposes a novel machine learning-based MCDA framework that combines an additive model and a fully-connected multilayer perceptron (MLP) to achieve good performance while capturing the explicit relationships between individual attributes and the prediction. The additive model uses marginal value functions to approximate the explicit relationship between the outcome and individual attributes whereas the MLP is used to capture the implicit high-order correlations between attributes in the model. We estimate the parameters in the model under a neural network framework that automatically balances the trade-off between two components. Note that we use MCDA approaches as descriptive models to represent the preferences that have already been expressed, rather than to help the DM to construct the preferences.

We validate the proposed model using a set of synthetic datasets and three real datasets. Specifically, the simulation experiments respectively show the impact of predefined parameters on model performance when data is either extremely complex or simple. We explain the obtained model and compare its performance with that of baseline interpretable models, i.e., GAM, logistic regression models, a conventional MCDA method, an MLP model, and a recent interpretable model in the literature.

The contributions of this study are as follows. First, we propose a hybrid model that integrates MCDA approaches with a neural network. It extends the conventional MCDA to quantify the explicit contribution of individual attributes to the outcome in the presence of the implicit high-order interactions and nonlinearities. The experimental results demonstrate that the proposed model outperforms conventional MCDA approaches while maintaining good interpretability. Second, to the best of our knowledge, this paper is the first pilot work that introduces the value function-based preference disaggregation approaches of MCDA to the machine learning models to enhance the model interpretability. Third, the proposed framework is flexible and extensible, especially the nonlinear part, which can be modified or replaced by other network structures or modeling schemes for various tasks.

The rest of the paper is organized as follows. In Section 2, we introduce the framework for the proposed model. The simulation and real case experiments are presented in Section 3 and some discussions about the proposed framework are provided in Section 4. We conclude the paper in Section 5.

2. Framework for the interpretable model

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_1^N$ be the dataset of size N , $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,n})^T$ be the i th attribute vector with n attributes², and y_i be the target/response value. In this study, we consider a binary classification problem where $y_i \in \{0, 1\}$. The proposed framework can be easily extended to multi-classification and regression problems.

2.1. The additive model

The value function-based preference disaggregation approaches of MCDA assume that for each attribute vector $\mathbf{x}_i \subseteq \mathbb{R}^n$, there is a global value function in the following form:

$$\begin{aligned} F(\mathbf{x}_i) &= w_1 \cdot v_1(x_{i,1}) + w_2 \cdot v_2(x_{i,2}) + \dots + w_n \cdot v_n(x_{i,n}) \\ &= \sum_{j=1}^n w_j v_j(x_{i,j}) \end{aligned} \quad (1)$$

² In MCDA, $\mathbf{x}_i$ is called an alternative with n criteria/attributes.

where $0 \leq w_j \leq 1, j = \{1, 2, \dots, n\}$ represents the importance of the j th attribute and $v_j(\cdot)$ is a marginal value function. Note that we reply the shape and positive/negative effect of the marginal value function to capture the contribution of individual attributes. Thus we set the weight w_j to be positive to represent the relative importance of the j th attribute, which can positively or negatively affect the global value. The *global value function* $F(\cdot)$ linearly sums contributions of individual attributes.

Although the global value function is in an additive form, the marginal value functions themselves can be in any form, often nonlinear. In the literature, the marginal value function can be in a simple linear (weighted sum) form [24], monotonic and non-monotonic piecewise linear forms [10,25,26], and monotonic polynomial form [13]. In this paper, we allow the marginal value functions to be in a non-monotonic polynomial form. The motivations are derived from two facets. First, the polynomial form enhances the expressiveness of the preference model to capture non-monotonic preferences. The piecewise linear or monotonic polynomial functions fail to capture all information in a larger learning set [13]. The nonlinearity and non-monotonicity of polynomial forms can better fit the complex relationships between attributes and the outcome, leading to better performance. Second, it is critical to examine the marginal rates of substitution in economics and management studies while analyzing human behavior. A non-derivative value function, for instance the piecewise linear functions, cannot capture the inflexion point where the marginal rate of substitution grows or diminishes more quickly. Such a model can exploit human behaviors, and thus is more convincing and has more managerial meaning for the DM in management scenarios.

To capture the first-order (e.g. monotonicity) and second-order (e.g. marginal rate in substitution) derivative patterns of the attributes' contributions to the prediction, we extend and generalize state-of-the-art MCDA models to allow the j th marginal value function to be in a smooth and non-monotonic form of D_j degrees:

$$v_j(x_{i,j}) = p_{j,1}x_{i,j}^1 + p_{j,2}x_{i,j}^2 + \dots + p_{j,D_j}x_{i,j}^{D_j} \quad (2)$$

where $p_{j,d} \in \mathbb{R}, d = \{1, 2, \dots, D_j\}$ is the coefficient of the d th degree and D_j is the highest order of degree on the j th attribute.

2.2. Neural network-based MCDA

An additive model like Eq. (1) does not model any interactions between attributes. Therefore, we propose a novel model, namely the **Neural Network-based Multiple Criteria Decision Aiding** (NN-MCDA), in the following form:

$$U(\mathbf{x}_i) = \alpha \sum_{j=1}^n w_j v_j(x_{i,j}) + (1 - \alpha) f(x_{i,1}, x_{i,2}, \dots, x_{i,n}) \quad (3)$$

$$= \alpha F(\mathbf{x}_i) + (1 - \alpha) f(\mathbf{x}_i) \quad (4)$$

where $U(\mathbf{x}_i)$ is the *global score* of $\mathbf{x}_i$, $f(\cdot)$ is a latent function of all attributes, and $\alpha \in [0, 1]$ is a trade-off coefficient. If $U(\cdot)$ directly outputs the values on the right side, Eq. (4) describes a regression problem. If $U(\cdot)$ is the logistic function that transforms original output into $[0,1]$ describing the probability of $\mathbf{x}_i$ being assigned to a positive class, Eq. (4) represents a classification problem. $f(\cdot)$ is used to capture the high-order interrelationships between attributes in the model. We can use any complexity models to fit $f(\cdot)$ for better performance, for instance we use a MLP in this paper [27]. While using an MLP form of $f(\cdot)$, it is not transparent, meaning that we do not know the exact structure of $f(\cdot)$. $F(\cdot)$ captures the explainable form of the marginal value functions. The non-transparent $f(\cdot)$ describes the complex patterns that are not readily useful to the DM. Coefficient α balances the trade-off between $F(\cdot)$ and $f(\cdot)$. If α is close to 1, the model tends to be in a

simple additive form. If α is close to 0, we obtain a full complexity model.

The utilized joint training process is shown in Fig. 1. The original input values $\mathbf{x}_i = (x_{i,1}^1, \dots, x_{i,j}^1, \dots, x_{i,n}^1)$ are in the 1-degree polynomial form. Before feeding it into the neural network, we transform the original input vector into the form that corresponds to the polynomial form, i.e., $\Phi(\mathbf{x}_i) = (x_{i,1}^1, \dots, x_{i,1}^{D_1}, x_{i,2}^1, \dots, x_{i,2}^{D_2}, \dots, x_{i,n}^1, \dots, x_{i,n}^{D_n})^T$. In the input layer, a single-layer network without any activation functions is provided to reconstruct Eq. (1). It has $\sum_{j=1}^n D_j$ units and the weight for each unit corresponds to a particular $p_{j,d}$. We denote the output of the linear component with z_i^{linear} , and

$$z_i^{linear} = (w_1, \dots, w_j, \dots, w_n) \begin{pmatrix} \mathbf{p}_1^T \phi(x_{i,1}) \\ \vdots \\ \mathbf{p}_j^T \phi(x_{i,j}) \\ \vdots \\ \mathbf{p}_n^T \phi(x_{i,n}) \end{pmatrix} = \mathbf{w}^T \mathbf{P}_i \quad (5)$$

where $\mathbf{p}_j = (p_{j,1}, p_{j,2}, \dots, p_{j,D_j})^T$ is the vector of coefficients in the j th polynomial marginal value function, $\phi(x_{i,j}) = (x_{i,j}^1, \dots, x_{i,j}^{D_j})^T$ is the transformed input vector, $\mathbf{w}$ is the vector of weights of attributes, and $\mathbf{P}_i$ contains marginal values of i th attribute vector. Note that, Eq. (5) is actually a specific case of Eq. (1). In Eq. (1), the marginal value functions $v(\cdot)$ can be in any shapes (e.g. piecewise linear). However, in this study, we allow them to be in a non-monotonic polynomial form in Eq. (2). Note that z_i^{linear} is a special form of $F(\cdot)$.

The nonlinear component is a standard MLP. It is used to learn high-order correlations between attributes. Similarly, by summing every D_j units we can obtain a marginal value on the j th attribute. For activation functions, we opt for Rectifier (ReLU) because it has been proved to be biologically plausible and non-saturated, and could avoid the over-fitting problem [28]. We also tried other forms of activation functions (Sigmoid and TanH), and ReLU indeed led to better performance. An L -layer MLP is defined as:

$$\begin{aligned} \mathbf{z}_1(\mathbf{x}_i) &= \Phi(\mathbf{x}_i), \\ \mathbf{z}_2(\mathbf{x}_i) &= a_1(\mathbf{W}_1^T \mathbf{z}_1 + \mathbf{b}_1), \\ &\dots \end{aligned} \quad (6)$$

$$\begin{aligned} \mathbf{z}_L(\mathbf{x}_i) &= a_{L-1}(\mathbf{W}_{L-1}^T \mathbf{z}_{L-1} + \mathbf{b}_{L-1}), \\ z_i^{nonlinear} &= \mathbf{h}^T \mathbf{z}_L(\mathbf{x}_i), \end{aligned}$$

where $\mathbf{W}_l$, $\mathbf{b}_l$, and $a_l(\cdot)$ denote the weight matrix, bias vector, and activation function for the l th layer, respectively, and $\mathbf{h}$ is the final weight vector at the last layer. The input of the MLP model is the same as the input for the linear part, i.e., $\Phi(\mathbf{x}_i)$.

The output is the probability of $y_i = 1$, we have

$$P(\hat{y}_i = 1 | \mathbf{x}_i) = \sigma(\alpha z_i^{linear} + (1 - \alpha) z_i^{nonlinear}) \quad (7)$$

where $\sigma(\cdot)$ is a sigmoid function.

To estimate the parameters, we minimize the mean square error (MSE):

$$MSE = \frac{1}{N} \sum_{i=1}^N (P(\hat{y}_i = 1 | \mathbf{x}_i) - y_i)^2 \quad (8)$$

We can adopt a variety of optimization methods to minimize Eq. (8), such as Stochastic Gradient Descent (SGD), Adaptive Gradient Algorithm (Adagrad) and Adaptive Moment Estimation (Adam). Please refer to [29] for details of the optimization procedure.

The interpretability of the model refers to the capacity in developing marginal value functions, which capture the relationship between individual attributes and prediction. With the proposed

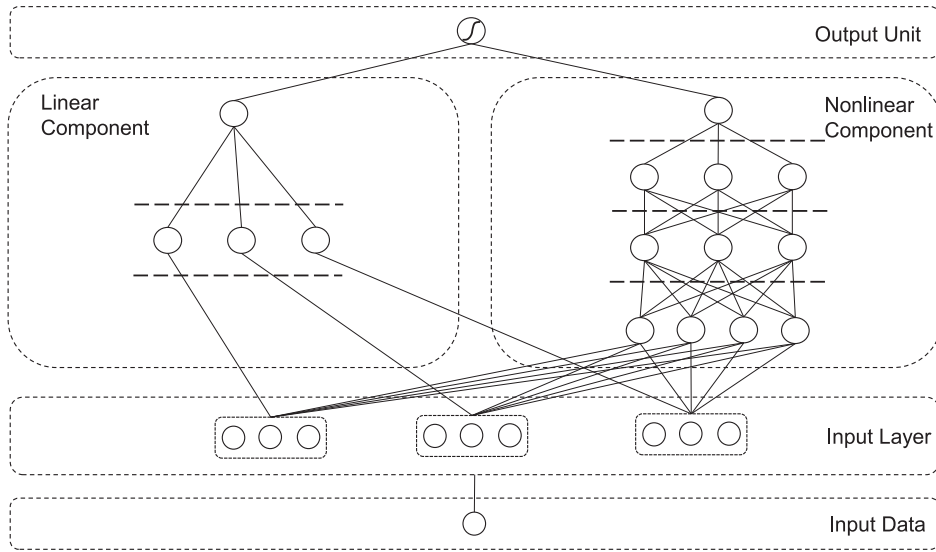


Fig. 1. The framework for NN-MCDA.

model, the DM can know what attributes are more important for the prediction, what values of an attribute are positively or negatively associated to the prediction, and where the convexity and concavity of the function are changed.

Proposition 1. When $a_l(\cdot), l = 1, \dots, L$ are linear activation functions, the proposed NN-MCDA model degenerates to an additive model.

Proof. See Appendix A $\square$

Proposition 2. When $a_l(\cdot)$ is a nonlinear activation function, if $L = 1$, the lower-order interacting attributes can be explicitly expressed. If $L \geq 2$, the higher-order interacting attributes cannot be explicitly expressed.

Proof. See Appendix B

Given Propositions 1 and 2, a neural network with nonlinear activation functions and more than one layer will bring high-order attribute interactions and complex attribute transformations, which are difficult to be described by an explicit function. The presence of these nonlinearities affects the main effects of the individual attributes. Therefore, the proposed NN-MCDA aims to learn the main effects in the presence of possible attribute interactions and their transformations instead of exploring the explicit form of these nonlinearities. $\square$

2.3. Application to multiple criteria ranking problems

In this subsection, we will show how to apply NN-MCDA to traditional multiple criteria ranking problems where alternatives are ranked based on the DM's judgment. In this paper, alternatives are represented as attribute vectors.

Let $\mathbf{x}_i \succeq \mathbf{x}_k$ denote that an attribute vector $\mathbf{x}_i$ is at least as good as $\mathbf{x}_k$, and $\mathbf{x}_i > \mathbf{x}_k$ denote that $\mathbf{x}_i$ is better than $\mathbf{x}_k$. Note that the symbol ' $\succeq$ ' (or ' $>$ ') does not necessarily require that each element in $\mathbf{x}_i$ is at least as good as (or better than) that in $\mathbf{x}_k$. It actually indicates that one alternative is at least as good as (or better than) another one based on the DM's judgment. For each pair $(\mathbf{x}_i, \mathbf{x}_k)$, we define $y_{i,k}$ as follows:

$$y_{i,k} = \begin{cases} 1, & \text{if } \mathbf{x}_i \succeq \mathbf{x}_k, \\ 0, & \text{if } \mathbf{x}_k > \mathbf{x}_i, \end{cases} \quad (9)$$

and the difference between global scores of $\mathbf{x}_i$ and $\mathbf{x}_k$ is:

$$\begin{aligned} U(\mathbf{x}_i) - U(\mathbf{x}_k) &= \alpha \sum_{j=1}^n w_j v_j(x_{i,j}) + (1 - \alpha) f(\mathbf{x}_i) \\ &\quad - [\alpha \sum_{j=1}^n w_j v_j(x_{k,j}) + (1 - \alpha) f(\mathbf{x}_k)] \\ &= \alpha \sum_{j=1}^n w_j (v_j(x_{i,j}) - v_j(x_{k,j})) + (1 - \alpha) [f(\mathbf{x}_i) - f(\mathbf{x}_k)] \\ &= \alpha \sum_{j=1}^n w_j \sum_{d=1}^{D_j} p_{j,d} (x_{i,j}^d - x_{k,j}^d) + (1 - \alpha) [f(\mathbf{x}_i) - f(\mathbf{x}_k)] \\ &= \alpha \sum_{j=1}^n w_j (\mathbf{p}_j^T \times \phi(x_{i,j}, x_{k,j})) + (1 - \alpha) \Theta(\Phi(\mathbf{x}_i, \mathbf{x}_k)) \\ &= \alpha \mathbf{w}^T (\mathbf{P}_i - \mathbf{P}_k) + (1 - \alpha) \Theta(\Phi(\mathbf{x}_i, \mathbf{x}_k)) \end{aligned}$$

where $\phi(x_{i,j}, x_{k,j}) = (x_{i,j}^1 - x_{k,j}^1, x_{i,j}^2 - x_{k,j}^2, \dots, x_{i,j}^{D_j} - x_{k,j}^{D_j})^T$. Let $\Phi(\mathbf{x}_i, \mathbf{x}_k)$ be the aggregated vector for $\phi(x_{i,j}, x_{k,j})$:

$$\Phi(\mathbf{x}_i, \mathbf{x}_k) = \left(\underbrace{x_{i,1}^1 - x_{k,1}^1, \dots, x_{i,1}^{D_1} - x_{k,1}^{D_1}}_{D_1}, \underbrace{x_{i,2}^1 - x_{k,2}^1, \dots, x_{i,2}^{D_2} - x_{k,2}^{D_2}}_{D_2}, \right. \\ \left. \underbrace{x_{i,3}^1 - x_{k,3}^1, \dots, x_{i,3}^{D_3} - x_{k,3}^{D_3}}_{D_3}, \dots, \underbrace{x_{i,n}^1 - x_{k,n}^1, \dots, x_{i,n}^{D_n} - x_{k,n}^{D_n}}_{D_n} \right)^T$$

and $\Theta(\Phi(\mathbf{x}_i, \mathbf{x}_k))$ be a function of $\Phi(\mathbf{x}_i, \mathbf{x}_k)$. We fit $\Theta(\cdot)$ function to approximate the value of $f(\mathbf{x}_i) - f(\mathbf{x}_k)$. Note that in some decision problems, the attribute weights $w_j, j = 1, \dots, n$ in Eq. (3) are normalized to $[0,1]$ and $\sum_{j=1}^n w_j = 1$, which are useful for interpreting the trade-offs between attributes³ To address this issue, we apply the following transformation:

³ Note that the trade-off between attributes is similar to attribute importance, but the trade-off emphasizes that assigning more weight to an attribute would decrease other attributes. That usually leads to a situation where some attributes have almost no effects on the predictions, which is unexpected because the selected attributes are often summarized based on DM's prior knowledge and their requirements. In this regard, we tend to train our model without normalization

- For each attribute $g_j, j = 1, \dots, n$, the normalized weight is $w'_j = \frac{w_j}{\sum_{j=1}^n w_j}$.
- The new global score is $U'(\mathbf{x}_i) = \alpha \sum_{j=1}^n w'_j v_j(x_{i,j}) + \frac{(1-\alpha)}{\sum_{j=1}^n w_j} f(x_{i,1}, x_{i,2}, \dots, x_{i,n})$. Moreover, the ordinal relations among all attribute vectors are preserved since $U'(\mathbf{x}_i) - U'(\mathbf{x}_k) = \frac{U(\mathbf{x}_i) - U(\mathbf{x}_k)}{\sum_{j=1}^n w_j}$ and $U(\mathbf{x}_i) \geq U(\mathbf{x}_k) \Leftrightarrow U'(\mathbf{x}_i) \geq U'(\mathbf{x}_k)$.

Given the input data $\mathcal{D} = \{(\Phi(\mathbf{x}_i, \mathbf{x}_k), y_{i,k})\}$, instead of mathematical programming, we can now use the machine learning scheme in Section 2.2 to infer the preference model and rank other attribute vectors. The output $\hat{y}_{i,k} = \sigma(U(\mathbf{x}_i) - U(\mathbf{x}_k))$ is the probability that $\mathbf{x}_i$ is at least as good as $\mathbf{x}_k$. We can predefine two thresholds η^1 and η^2 , where $0 \leq \eta^1 \leq \eta^2 \leq 1$. If $\eta^2 \leq \hat{y}_{i,k}$, then $\mathbf{x}_i > \mathbf{x}_k$, and if $\eta^1 \leq \hat{y}_{i,k} \leq \eta^2$, then $\mathbf{x}_i \sim \mathbf{x}_k$ and otherwise, $\mathbf{x}_k > \mathbf{x}_i$. If we use the normalized weights, since the probability $\hat{y}'_{i,k} = \sigma(U'(\mathbf{x}_i) - U'(\mathbf{x}_k))$ is transformed nonlinearly, the predefined thresholds should also be transformed as follows, $\eta^1_{i,k} = \frac{\hat{y}'_{i,k}}{\hat{y}_{i,k}} \eta^1$, $\eta^2_{i,k} = \frac{\hat{y}'_{i,k}}{\hat{y}_{i,k}} \eta^2$, to preserve the ordinal relations. In this way, the traditional multiple criteria ranking approaches can handle larger datasets and obtain smoother and more flexible marginal value functions to assist the DM. We present the simulation results in Section 3.1 and the results using real datasets in Section 3.2.

2.4. Usefulness of the proposed framework

The usefulness of the proposed framework can be summarized from two facets. From the perspective of extending MCDA approaches, NN-MCDA assumes the DM's preference is described by polynomial value functions and generalizes the MCDA approaches by allowing all attributes to be possibly non-monotonic and interacting. The nonlinear component in NN-MCDA can capture the implicit high-order interactions between attributes and their complex nonlinear transformations. Therefore, the polynomial marginal value functions are more expressive than monotonic piece-wise linear function [13] and are learned in the presence of interacting attributes. Moreover, with the help of machine learning models, instead of relaxing the relevant constraints, NN-MCDA uses standard gradient-based algorithms to optimize the parameters in the neural network to determine the characteristics of the marginal value functions. Comparing to the conventional MCDA models, the proposed NN-MCDA is more expressive and accurate and is able to efficiently handle large datasets.

From another perspective, NN-MCDA, unlike simple machine learning-based models, allows the DM to gain insights into the characteristics of the decision process and the interpretations of the results. Comparing with the static coefficient obtained from regression models, the polynomial value functions are more expressive and easier to be interpreted by the DM:

First, the attribute importance usually has a long-tail distribution, with a few of them being very important and the majority of them being less important [23]. The characteristics of the marginal value functions can reveal the importance of the corresponding attribute. If a marginal value function is close to 0 for the whole scale of the attribute values, it indicates that the attribute is either not important to the DM or the characteristic of the marginal value function is wrongly captured, because the change of this attribute has little influence on the predictions. When this is the case, we need to interact with the DM to determine whether we preserve this attribute or calibrate the model. In this regard, the proposed framework can perform model selection and modification (similar

to statistical approaches like LASSO). For example, while predicting if a patient has the flu, the marginal value function of "room humidity" is in a shape like the marginal value function 1 in Fig. 2, it is possible that "room humidity" has little contribution to the flu. However, whether we abandon it should be determined by a physician.

Second, the increasing and decreasing tendencies of the marginal value function curves reveal the change of the DM's non-monotonic preference. We focus on the monotonicity inflexion points because they can determine to what attribute values the DM is more sensitive. Moreover, if we partition the marginal value function curve by these points, we can discretize the continuous attribute into smaller ranges in which the DM's preference is monotonic. Such smaller intervals are useful for personalization (e.g. customer segmentation) and strategy-making tasks in management. For example, while evaluating the company's performance, if the marginal value function of "cash to total assets ratio" is like the second function in Fig. 2, we can learn that a company with a very small or large "cash to total assets ratio" is in a bad condition. Companies with a large ratio are suggested to use the cash to do more investigations, whereas companies with a small ratio are suggested to save general expenses so that more cash can be used in new investigations.

Third, since the marginal value function returns a "score" that is added to the global value, it is crucial to determine whether the attribute positively or negatively contributes to the outcome. If a marginal value function is above/below zero, the corresponding attribute is positively/negatively associated with the prediction. The marginal value function can capture the sign change (if any) of an attribute's contribution and provide the DM an exact attribute value where the sign changes. This is more informative than the statistical models that only provide a fixed coefficient representing the either positive or negative effect of the attribute. For example, when predicting the risk for depression among adults, the marginal value function of "age" may have a shape similar to the third function in Fig. 2 (please also refer to Fig. 17, which is drawn from the real data). The shape of this curve indicates that the risk for depression does not increase while aging if the adult is younger than a threshold. The risk will increase once the adult is older than that threshold (the threshold is 71.58 in the real data introduced in Section 3.3). Statistical models, on the other hand, can only conclude that age has either a negative or positive effect on the depression risk. We need to segment the adults to pre-defined age groups to capture such a sign change effect.

Fourth, the concavity (and convexity) of the marginal value function can directly reflect the changing rate of the DM's preference. Such information is important to both economics and marketing problems. For example, if the consumer's preference to "discount rate" is in the same shape of the marginal value function 4 in Fig. 2, it implies that at the beginning, along with the increase of the discount rate, the consumer's utility (propensity to consume the product) grows more quickly. However, when the discount rate is over a specific value, it gives a signal that the product is possibly of bad quality. Although the consumer's utility still grows, its rate of increase starts to slow down. This provides the DM with a conclusion that keeping the discount rate at a medium level could maximize the profit.

3. Experiments

To validate the proposed NN-MCDA model, we perform experiments with both synthetic and real datasets. We use *area under the curve* (AUC) of *receiver operating characteristic* (ROC) curve to measure the model performance. We recall the definition of AUC in Appendix C. In Section 3.1, three simulation experiments examine (a) the influence of the degree of a polynomial on the predic-

but provide normalized weights to evaluate the trade-offs between attributes [9]. Moreover, there are few minor differences on performances when using normalized weights or not.

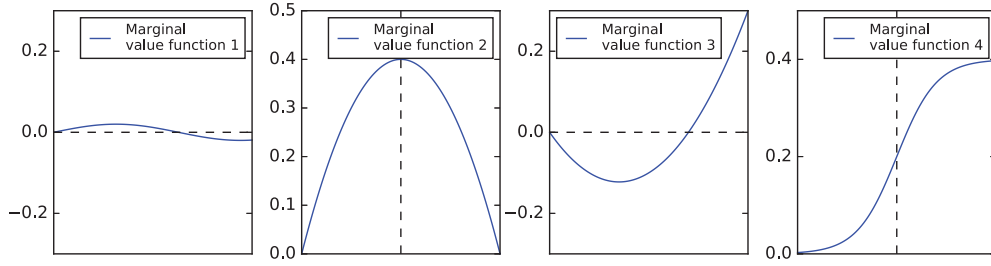


Fig. 2. Exemplary marginal value functions.

tion performance, (b) the influence of the value of α , the trade-off coefficient, on prediction performance, and (c) the goodness of the proposed NN-MCDA approach in fitting the given marginal value functions. In Section 3.2, we first apply the NN-MCDA model to a multiple criteria decision problem where we rank universities based on the employer reputation. Then we predict the risk for geriatric depression with useful interpretations of the risk factors with a higher resolution. At last, we perform the NN-MCDA on open-access data to compare the results with a published paper.

Note that NN-MCDA uses a gradient-based algorithm to optimize the parameters in the neural network. The solutions are local optimal so that the obtained marginal value functions may be different in each trial. In the experiments with real-world data, we train the model for several trials. In each trial, the parameter initialization follows the same Gaussian distribution. The final marginal value functions are determined by averaging the optimal parameters in different trials. In each experiment, we repeat this learning process to check the robustness of the results obtained from multiple sets of trials. Although there are minor differences among the marginal value functions, the learning processes provide the same basic shapes, for instance the monotonicity.

3.1. Simulations

For brevity, we set equal predefined degrees of polynomial for all marginal value functions in the subsequent experiments. We generate three typical synthetic datasets (from the simplest to very complex) as follows:

1. Uniformly draw N attribute vectors with n attributes whose values are within $[0,1]$.
2. We generate three datasets. (a) For the first dataset $\mathcal{D}_1^n$, all n attributes have equal importance and the actual marginal value functions are identity functions. The global score for each attribute vector is a linear summation of n attribute values without any attribute interactions and an additional noise term that is in a standard normal distribution; (b) The second dataset $\mathcal{D}_{\text{polynomial-3}}^n$ randomly generates 3-degree polynomials marginal value functions for n attributes, and the global score is the summation of marginal values, all $\binom{n}{2}$ attribute interactions and a standard normal noise term. (c) The third dataset $\mathcal{D}_{\text{polynomial-15}}^n$ is extremely complex. The global score is the summation of n 15-degree polynomial marginal values, some attribute interactions (pairwise, triple-wise) and a standard normal noise term⁴.

3. We compare global scores between each pair of attribute vectors. If $U(\mathbf{x}_i) - U(\mathbf{x}_k) \geq 0$, then $y_{i,k} = 1$, otherwise, $y_{i,k} = 0$. Note that the actual input is the transformed attribute vector.

3.1.1. Experiment i: Relationship between degree of polynomial and model performance

The first simulated experiment aims at exploring the relationship between the predefined degree of polynomial and AUC. The parameters used in the experiment are shown in Table 1. For each setting, we iteratively repeat the experiment 10 times and record the averaged AUC. In this experiment, the numbers of iterations are determined using fivefold cross-validation: We partition the training set into five sets and set aside one of them as a validation set. We then train the model using the other four partitions and use the validation set to check the convergence. This procedure is repeated five times and the averaged number of iterations is used to train the final model with the whole dataset.

Figs. 3, 4 and 5 report the averaged AUC for the testing set with different training sizes using the three synthetic datasets. Though there are no obvious relationships between the training sizes and the model performance, we find two interesting patterns. First, higher predefined degrees of polynomials can lead to higher accuracy when convergence. That results from the ability of the underlying model to capture more complicated nonlinearity. However, higher degrees of polynomials usually require more iterations to converge. More specifically, we depict the averaged computational time for each training process in Fig. 6. Apparently, while increasing the model complexity, for example, using higher degrees of the polynomial marginal value function and considering more attributes, the average computational time to converge also increases almost linearly. Another interesting pattern is that the shape of the AUC curves (Figs. 3, 4 and 5) can fit a concave function in general. When the degree increases, the AUC improvement (over the model with the immediate smaller degree) is becoming smaller. For example, the improvement is more obvious if we change the predefined degree from 1 to 3 than that if we change from 3 to 5 and from 5 to 10. The improvement diminishes quickly along with the increase of the predefined degree of polynomials. The greatest AUC improvement happens if we increase the degree to 3, while the improvement resulted from further increasing the degree to 5 and 10 is slim. The results suggest that it is not necessary to set a very large D_j for the seek of minor improvement because the computational cost increases much faster when D_j increases. Generally, we believe that a polynomial of degree 3 is sufficient to capture the characteristics of all three datasets. Higher degrees of polynomials have a risk for over-fitting and cost more computational time, but contribute little to accuracy.

We also compare the proposed NN-MCDA with baseline machine learning models and conventional MCDA models, including the standard MLP, polynomial linear regression (PLR) with 1, 2, 3, 5 and 10 degrees, GAM with 10 splines that are in 3 and 10 degrees of polynomials, SVMs all the three with linear, radial basis

⁴ To be specific, $\mathcal{D}_1^n$ considers an additive form $U(\mathbf{x}_i) = x_{i,1} + x_{i,2} + \dots + x_{i,n} + \epsilon$, where ϵ is a noise term. $\mathcal{D}_{\text{polynomial-3}}^n$ considers all pairwise interactions $U(\mathbf{x}_i) = f_1(x_{i,1}) + \dots + f_n(x_{i,n}) + g_{1,2}(x_{i,1}x_{i,2}) + \dots + g_{n-1,n}(x_{i,n-1}x_{i,n}) + \epsilon$, where $f_j(\cdot)$ is a random 3-degree polynomial function and $g_{j_1,j_2}(\cdot)$ is a random function of $x_{i,j_1}x_{i,j_2}$. Similarly, $\mathcal{D}_{\text{polynomial-15}}^n$ considers a function form $U(\mathbf{x}_i) = f_1(x_{i,1}) + \dots + f_n(x_{i,n}) + g_{1,2}(x_{i,1}x_{i,2}) + \dots + g_{n-1,n}(x_{i,n-1}x_{i,n}) + \dots + h_{j_1,j_2,j_3}(x_{i,j_1}x_{i,j_2}x_{i,j_3}) + \dots + \epsilon$, where $f_j(\cdot)$ is now a random 15-degree polynomial function, $g_{j_1,j_2}(\cdot)$ is a random function of $x_{i,j_1}x_{i,j_2}$ and $h_{j_1,j_2,j_3}(\cdot)$ is a random function of $x_{i,j_1}x_{i,j_2}x_{i,j_3}$.

Table 1
Parameter settings for simulation experiment I.

	N	# Comparisons	Training size	Predefined D_j	# Attributes
Setting	250	$\frac{250 \times 249}{2}$	0.5, 0.6, 0.7, 0.8, 0.9	1, 2, 3, 5, 10	3, 5

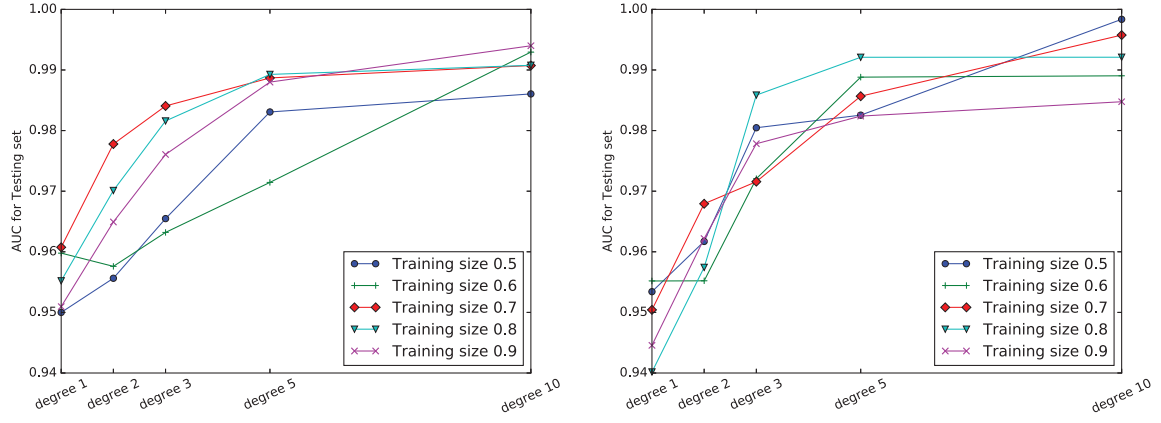


Fig. 3. The average AUC for the testing set using training set with different sizes from datasets D_1^3 and D_1^5 .

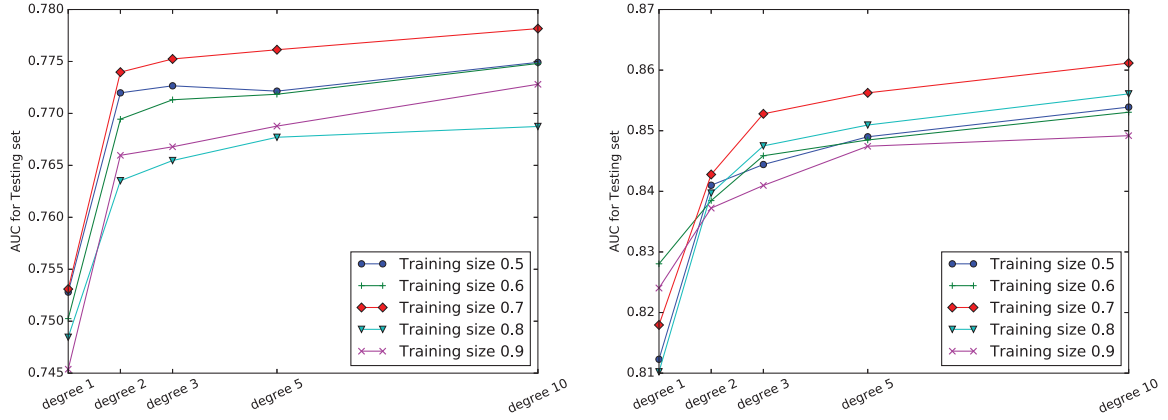


Fig. 4. The average AUC for the testing set using training set with different sizes from datasets $D_3^{\text{polynomial-3}}$ and $D_5^{\text{polynomial-3}}$.

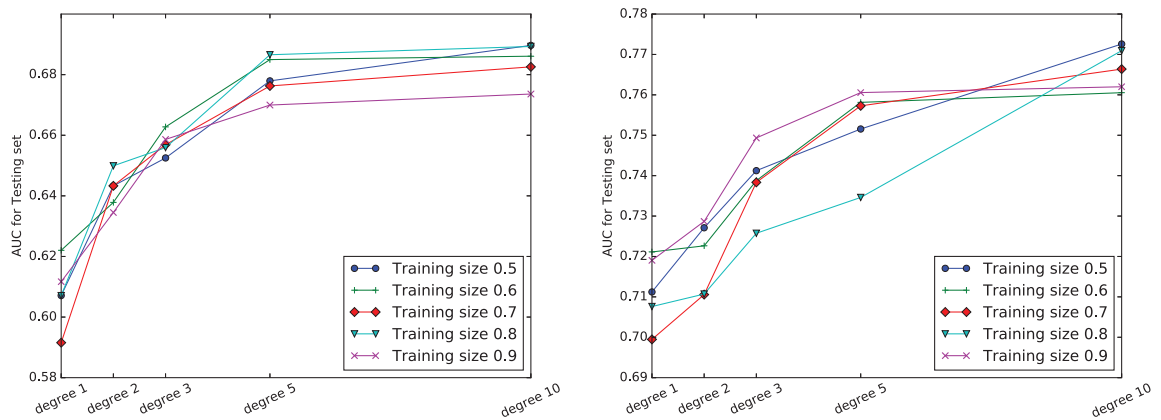


Fig. 5. average AUC for the testing set using training set with different sizes from datasets $D_3^{\text{polynomial-15}}$ and $D_5^{\text{polynomial-15}}$.

function (RBF) and polynomial kernels, single decision tree (DeciTr) models with 6, 10 and 20 maximum depths, and UTADIS with 5 and 10 equal sub-intervals [25]. The full results are presented in Appendix D. For the simplest dataset, both machine learning and interpretable models (including NN-MCDA and UTADIS) perform well. The performance drops rapidly when we use them to fit the

nonlinear and high-order datasets (shown in the last two Tables in Appendix D). Since the proposed NN-MCDA model and MLP can model the nonlinearity and attribute interactions, both of them achieve much higher AUCs as compared to the rest. As expected, although the performance of NN-MCDA is lower than MLP, the difference is relatively small. Both NN-MCDA and MLP outperform

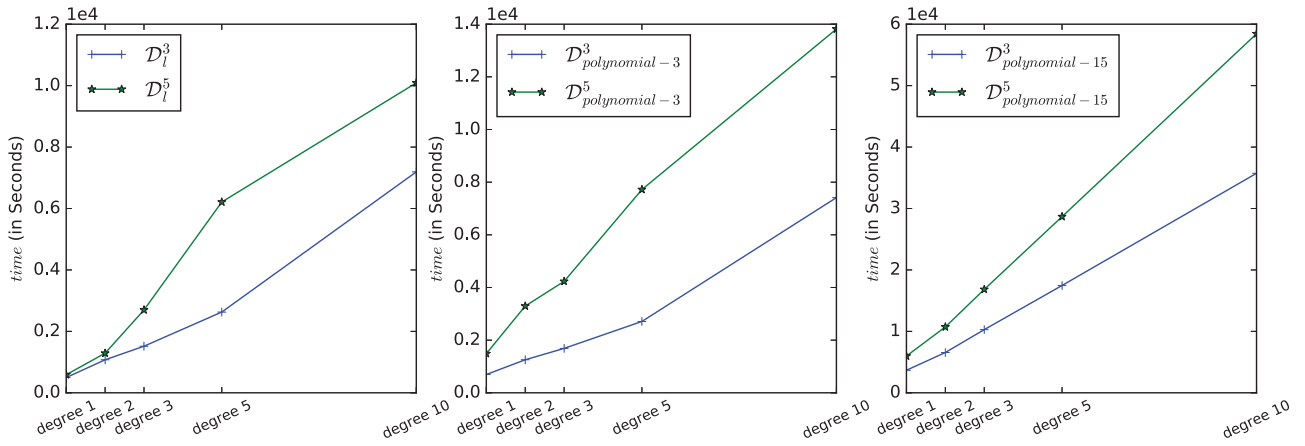


Fig. 6. The average computational time for different datasets. The recorded computational times in the figure are the averaged time for training all five training sizes given a predefined degree of polynomials.

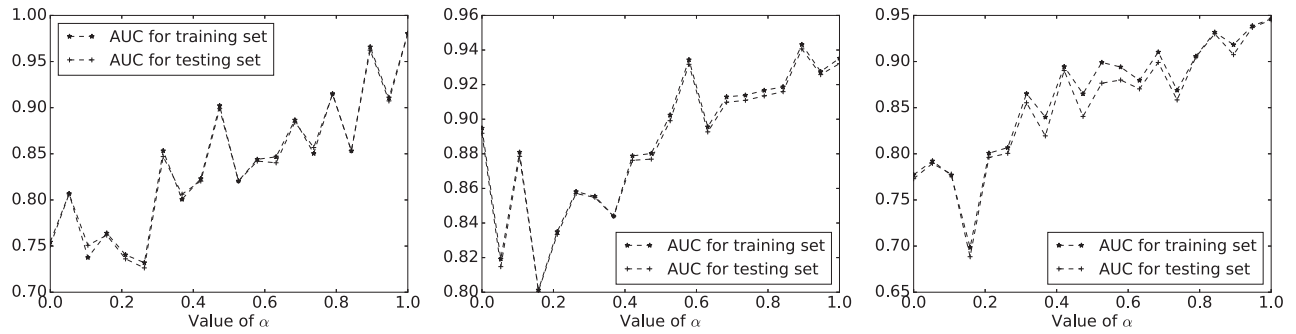


Fig. 7. From the left to right, the impact of α on AUC using $\mathcal{D}_l^3$, $\mathcal{D}_l^5$, $\mathcal{D}_l^{10}$, respectively.

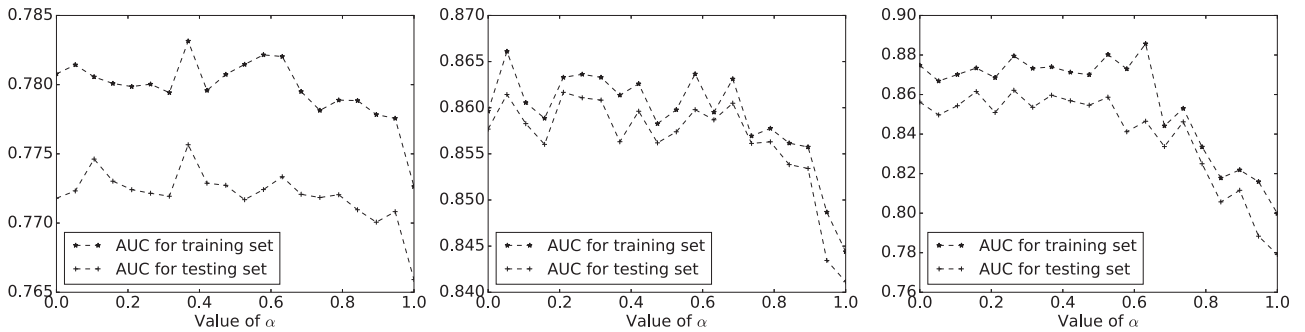


Fig. 8. From the left to right, the impact of α on AUC using $\mathcal{D}_{polynomial-3}^3$, $\mathcal{D}_{polynomial-3}^5$, $\mathcal{D}_{polynomial-3}^{10}$, respectively.

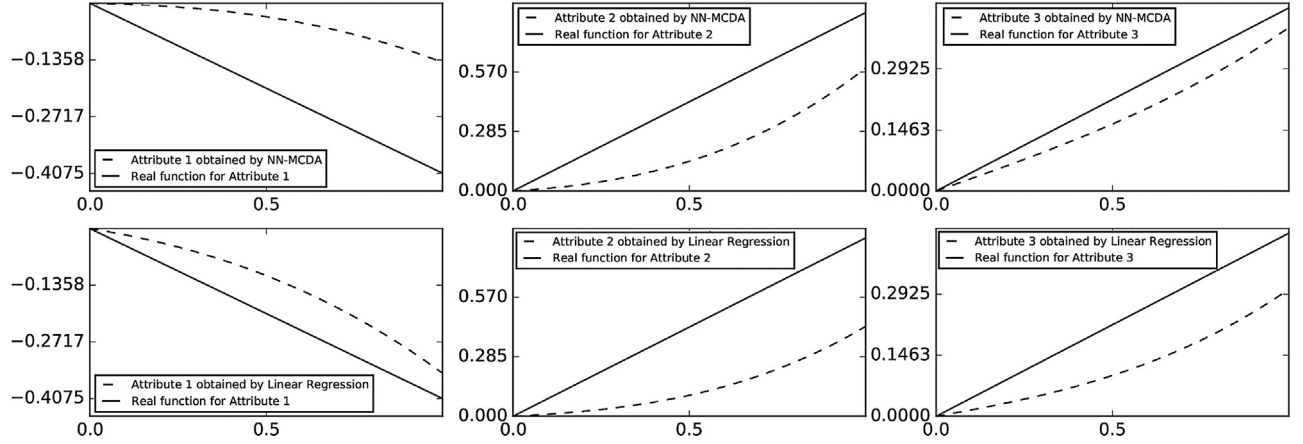
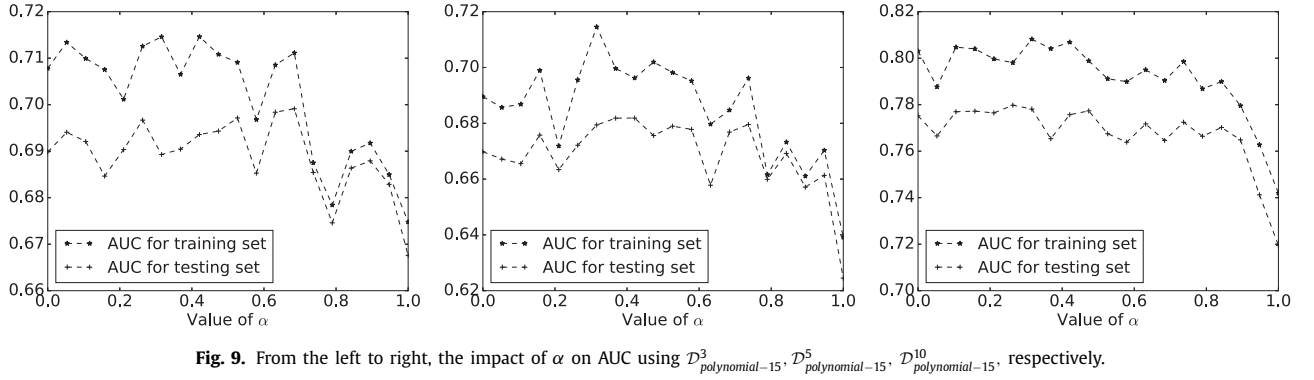
other baseline machine learning models and UTADIS significantly. More specifically, we can observe that GAM, SVM, and DeciTr have similar accuracy, which is higher than that of PLR, but lower than NN-MCDA. The performance of UTADIS is on a par with or better than PLR-D1 because the piecewise linear form is more expressive than simple linear form. However, both UTADIS with 5 and 10 sub-intervals perform worse than NN-MCDA. NN-MCDA's performance is close to the full complexity model, and at the same time, it preserves a certain degree of interpretability (as will be shown in the next experiment).

3.1.2. Experiment II: Impact of α on AUC

In this section, we focus on assessing how the performance of the NN-MCDA model is affected by the value of α , the weight for the linear component. We evenly sample 20 values within $[0, 1]$ as

the predefined α . For each fixed α , we train the NN-MCDA model using synthetic datasets introduced in the previous subsection. In this experiment, we use the SGD algorithm to optimize the parameters and the number of iterations is set as 250.

In Fig. 7 (results with the linearly generated datasets), though three curves have many monotonicity inflexions, there is a general trend that the greater the value of α , the better the performance. NN-MCDA obtains the best results when α is between 0.8 and 1.0 on these linearly generated datasets. In contrast, the AUC curves have a general decreasing trend when α increases for the non-linearly generated datasets (Figs. 8 and 9). In the extreme cases, where only the nonlinear MLP component (i.e., $\alpha = 0$) or the linear component (i.e., $\alpha = 1$) works, the model obtains the greatest or smallest average AUC.



Note that the three datasets have very different patterns. Datasets $\mathcal{D}^n$ use a set of simple linear marginal value functions whereas $\mathcal{D}^n_{\text{polynomial}-3}$ and $\mathcal{D}^n_{\text{polynomial}-15}$ simulate more complicated patterns. Theoretically, a full complexity model (MLP) can perfectly capture any patterns in the data at the cost of very large numbers of iterations and data samples for convergence. In practice, we often do not have sufficient data or computational time to achieve the optimal MLP solution. In this simulation experiment (31,125 data points and 250 iterations to fit the model), a pure MLP model (NN-MCDA with $\alpha = 0$) does not always lead to the best outcome. This result indicates that, in real-world managerial decision-making, a full complexity model is usually not the best one not only because of the lack of interpretability, but also the limited data and computational resources to optimize the model. It is sensible to allow the model to automatically adjust the trade-off coefficient α to avoid the scenarios where a very complex model is used to fit simple data, or a simple model is used to fit complex data.

3.1.3. Experiment III: Performance in fitting actual marginal value functions

This experiment studies the ability of the NN-MCDA model to reconstruct the actual marginal value functions. From Experiments I and II, we find that the NN-MCDA model with a degree equal to 3 has a good balance between prediction performance and computational cost. Therefore, in this experiment, we generate four typical synthetic models with different complexities. Each hypothetical model has three marginal value functions to estimate. Then, we use the synthetic models to generate datasets with the same attribute vectors (model input). We approximate the marginal value functions using an NN-MCDA model with a degree of 3. We also compare the obtained function with a baseline linear regression

model. The four synthetic models (from the simplest to extremely complicated) are described as follows:

- Synthetic model 1: Three linear marginal value functions. The global scores are the linear summation of three marginal values without interactions.
- Synthetic model 2: Three polynomial functions of degree 3. The global scores are the linear summation of three marginal values with all possible pairwise interactions.
- Synthetic model 3: A polynomial function of degree 15, a sigmoid function and an exponential function. The global scores are the linear summation of three marginal values without interactions.
- Synthetic model 4: The Model 3 with pairwise and triple-wise attribute interactions.

Figs. 10 to 13 reveal the actual and fitted marginal value functions obtained by the proposed NN-MCDA model and the baseline linear regression model. For a simple model with linear marginal value functions (Synthetic model 1), both the baseline linear regression model and the NN-MCDA can fit the actual functions well. Both models successfully capture the monotonicity of the original marginal value functions. This indicates that the NN-MCDA model is also applicable to simple prediction tasks that do not have attribute interactions or nonlinear associations between attributes and predictions.

When the attribute interactions are considered (Synthetic model 2), the proposed model outperforms the baseline linear regression model. In the first row of Fig. 11, the NN-MCDA model captures correct monotonicity changes of all three actual marginal value functions. Moreover, for the first and third attributes, it captures the concavity of the original functions. While the linear regression model gives opposite monotonicity of the actual functions

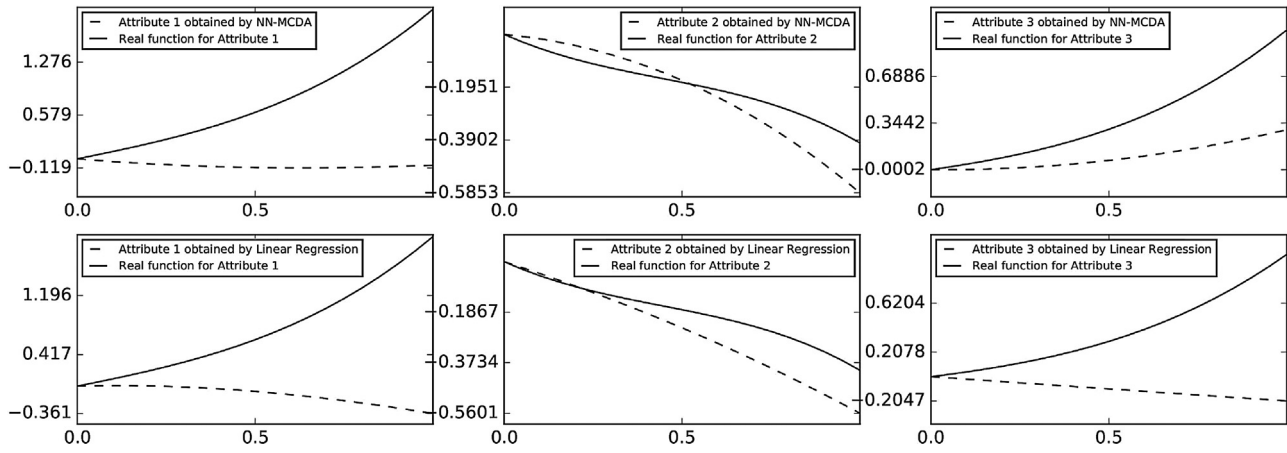


Fig. 11. The simulated and actual marginal value functions in model 2.

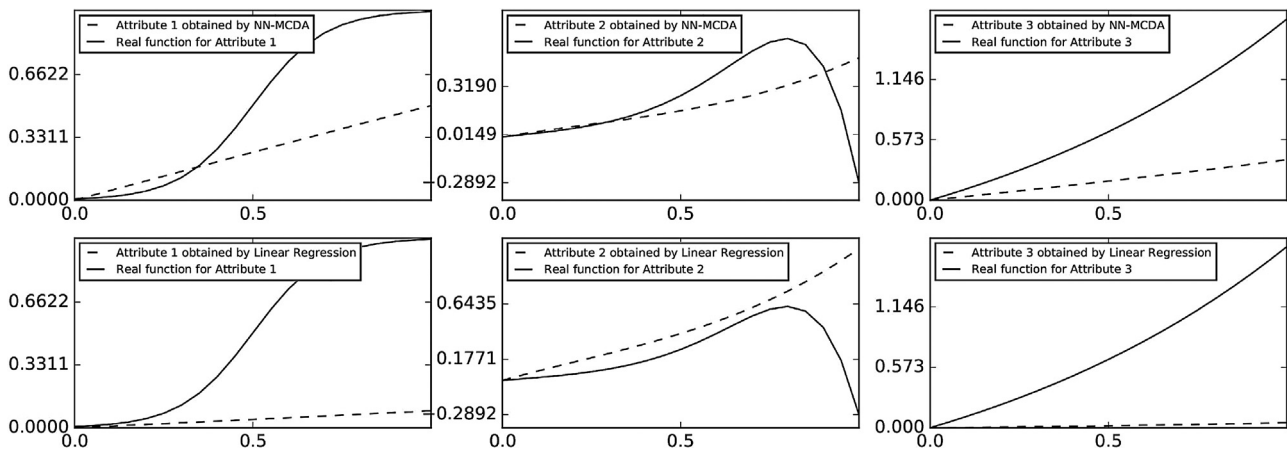


Fig. 12. The simulated and actual marginal value functions in model 3.

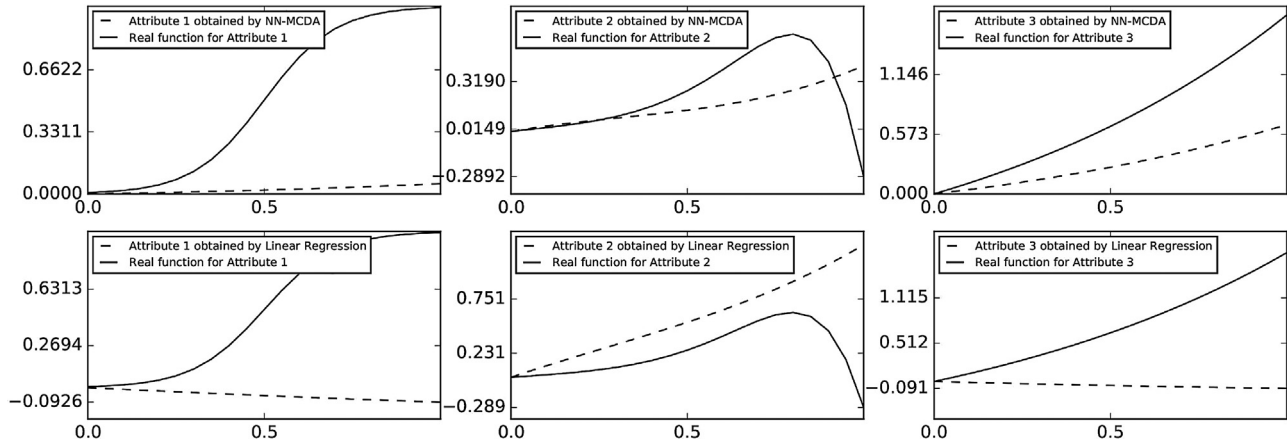


Fig. 13. The simulated and actual marginal value functions in model 4.

except for the second attribute. In addition, the linear regression model does not capture the concavity of the actual functions. The bad performance of the linear regression model is due to the fact that it can not capture the interactions between attributes, which often brings distorted interpretability.

For the model with more complex marginal value functions (Synthetic model 3), the linear regression model performs rather bad. In Fig. 12, the first and third attributes have a very negligible impact on the prediction in the fitted linear regression model. The NN-MCDA model, on the other hand, correctly captures the main

characteristics of the three attributes. Both models failed to capture the inflexion point for the second attribute. These results demonstrate that a low-degree NN-MCDA model (3-degree in this study) can fit both lower and higher degree marginal value functions, because the nonlinear component helps to deal with the complexities that cannot be captured by the predefined linear component. In Fig. 13, if the marginal value function is extremely complex (Synthetic model 4), though low-degree NN-MCDA cannot fully capture the characteristics of the marginal value functions, it still outperforms the baseline linear regression model. However, the real-

Table 2

The descriptive statistics of data. Original data is normalized into [0,1] in the following experiment.

	Mean	Std Dev	Minimum	Maximum
Employer-student connection	74.63	11.17	44.80	100.00
Alumni outcomes	55.25	16.74	26.40	100.00
Partnerships with employers	73.27	10.76	46.10	100.00
Graduate employment rate	77.88	7.93	54.00	100.00

world decision-making behaviors are usually not that complex (15-degree in Synthetics models 3 and 4). We will further validate the applicability and generalizability of the proposed NN-MCDA model with real datasets in the following section.

3.2. A multiple criteria ranking problem

QS world university ranking organization⁵ provides five carefully-chosen indicators to measure the universities' capacity in producing the most employable graduates, including employer reputation, employer-student connection, alumni outcomes, partnerships with employers and graduate employment rate. The metric of employer reputation, regarded as a key performance indicator, is based on over 40,000 responses to the QS Employer Survey. In this experiment, we apply the proposed NN-MCDA model to a multiple criteria ranking problem that predicts the employer reputation (human decision) using the other four quantitative indicators⁶. The descriptive statistics are shown in Table 2.

There are $\binom{250}{2}$ pairwise comparisons among 250 universities. To determine the predefined degree of polynomials, we respectively set D_j as 1, 2, and 3. We use the same fivefold cross-validation process to fit the model. We record the averaged AUC for both training and testing sets. We also select three baseline models, including a 3-layer MLP, a logistic regression model, a GAM, and UTADIS with 5 equal sub-intervals. The results of the average AUC are presented in Table 3. For each predefined degree, the full complexity model always obtains the best results whereas the logistic regression model and UTADIS perform the worst. Since NN-MCDA can model attribute interactions, it slightly outperforms the GAM.

Note that α is automatically determined by the learning process of the NN-MCDA. It represents a trade-off between the linear and nonlinear components in the NN-MCDA. Larger α means NN-MCDA gives more weight to the linear component. The results in Table 3 indicate that as we increase the degree of polynomials, α decreases, and model performance increases. This suggests that when we use a more complex model (as compared to 1-degree) to achieve better prediction performance, the linear component tends to be less important than the nonlinear component. The difference in α justifies the trade-off between model interpretability and performance.

⁵ It is an annual publication of university rankings by Quacquarelli Symonds (QS). <https://www.topuniversities.com>

⁶ Center for Epidemiologic Studies Depression (CES-D) scale is a self-report measure of the frequency of 20 depressive symptoms during the past week [31]. It is one of the most popular index assessing the risk for being depressed [32]. Age in years at the end of the survey is calculated from the respondent birth date and beginning survey date. The samples vary from 18 to 104 but very few of them are within 18–45 and 93–104 years [33]. The degree of education is measured by the years of getting an education. It is a categorical variable varying from 0 to 18. Note that for respondents whose degree of education is higher than 17 years, we set the degree of education as 18 [34]. Marital status is represented by the length of the longest marriage that the respondent ever had. It is a continuous variable and varies from 0 to 74.2 years [35]. Out-of-pocket expenditure refers to the expenses that the respondent pays directly to the health care provider without a third-party (insurer, or State). We consider the out-of-pocket medical expenditure in the previous 2 years [36]. Body mass index is the weight divided by the square height [30], which determines whether a respondent is overweight [37].

We depict the marginal value functions obtained by NN-MCDA, logistic regression model and GAM in Figs. 14, 15 and 16, respectively. In Figs. 14 and 15, the vertical axis is the individual attributes contributions to employer reputation. Specifically, for *employer-student connection*, NN-MCDA provides more plausible value functions than those by the linear regression. The shape of the curve indicates that the employer-student connection is positively associated with the university's reputation and such positive association becomes stronger along with the increase of the attribute value, and then decreases, but still positive. Such a curve indicates that although a university can enhance the reputation by increasing the value of employer-student connection, there exists an optimal value for this attribute that leads to the best efficiency. The first plot in Fig. 15 reveals a U-shape function and the main part of the function is under the baseline rate. It indicates that *employer-student connection* is negatively associated with the university reputation, which is counter-factual. For the attributes *alumni outcomes*, *partnerships with employers* and *graduate employment rate*, the value functions obtained by NN-MCDA and logistic regression model exhibit a monotonically increasing trend, which makes sense based on our common knowledge.

For the attribute *employer-student connection*, GAM obtains a generally increasing curve. Its shape is different from the curve captured by NN-MCDA. For the attributes *alumni outcomes* and *partnerships with employers*, GAM can obtain similar curves as NN-MCDA and logistic regression model, but their shapes are more complex. For the attribute *graduate employment rate*, GAM obtains unstable curves that are difficult to explain.

3.3. Predicting geriatric depression risk

Depression is a major cause of emotional suffering in later life. Geriatric depression reduces the quality of older adults' life and increases the risk of acquiring other diseases and committing suicide. The existing literature empirically studied the risk factors of geriatric depression but few of them gave insight into how these risk factors affect the prevalence of geriatric depression in detail (i.e.: the shape of the marginal value functions). It is more managerially helpful for clinical decision-making to prevent older adults from being depressed if we can understand how each risk factor influences the risk for depression at different scales.

The Health & Retirement Study (HRS) is a nationally representative longitudinal study of US adults [30]. It has been widely used in many medical studies because of its massive information about older adults' demographics, health status, health care utilization and costs, and other useful variables. We sample the data in 2014 ($N = 17,696$). In this experiment, given five predetermined attributes (risk factors) that are associated with geriatric depression, we want to capture the detailed effect of these risk factors at different scales, which are represented by marginal value functions. The descriptive statistics are described in Table 4⁷

⁷ Center for Epidemiologic Studies Depression (CES-D) scale is a self-report measure of the frequency of 20 depressive symptoms during the past week [31]. It is one of the most popular index assessing the risk for being depressed [32]. Age in years at the end of the survey is calculated from the respondent birth date and beginning survey date. The samples vary from 18 to 104 but very few of them are within 18–45 and 93–104 years [33]. The degree of education is measured by the years of getting an education. It is a categorical variable varying from 0 to 18. Note that for respondents whose degree of education is higher than 17 years, we set the degree of education as 18 [34]. Marital status is represented by the length of the longest marriage that the respondent ever had. It is a continuous variable and varies from 0 to 74.2 years [35]. Out-of-pocket expenditure refers to the expenses that the respondent pays directly to the health care provider without a third-party (insurer, or State). We consider the out-of-pocket medical expenditure in the previous 2 years [36]. Body mass index is the weight divided by the square height [30], which determines whether a respondent is overweight [37].

Table 3

Given polynomials of predefined degree, the averaged AUC for training and testing sets. The value in parentheses is α when convergence. There are minor differences between averaged AUC when predefined degrees are 2 and 3. Using polynomial of degree 2 is sufficient because a more complex model does not significantly increase AUC.

	NN-MCDA	Logistic regression	MLP	GAM (5 splines)	UTADIS-5
1-degree Training	0.66 (0.8988)	0.61 (1.00)	0.71(0.00)	0.64	0.64
1-degree Testing	0.64	0.57	0.70	0.63	0.63
2-degree Training	0.69 (0.4969)	0.64 (1.00)	0.73 (0.00)	0.67	–
2-degree Testing	0.68	0.63	0.71	0.66	–
3-degree Training	0.69 (0.4351)	0.65 (1.00)	0.74 (0.00)	0.68	–
3-degree Testing	0.68	0.65	0.73	0.66	–

Table 4

Descriptive statistics of variables used in experiment.

Outcome	Mean	Std Dev	Minimum	Maximum	Type
Center for Epidemiologic Studies Depression (CES-D)	1.52	2.05	0	8	Categ
Variables					
Age in years (AGE)	67.42	10.99	18	104	Cont
Degree of education (EDU)	12.83	3.44	0	18	Categ
Marital status (MS)	30.13	18.19	0	74.2	Cont
Out-of-pocket expenditure (OPE)	2,978.72	7,696.11	0	232,255.46	Cont
Body mass index (BMI)	28.24	7.14	0	76.6	Cont

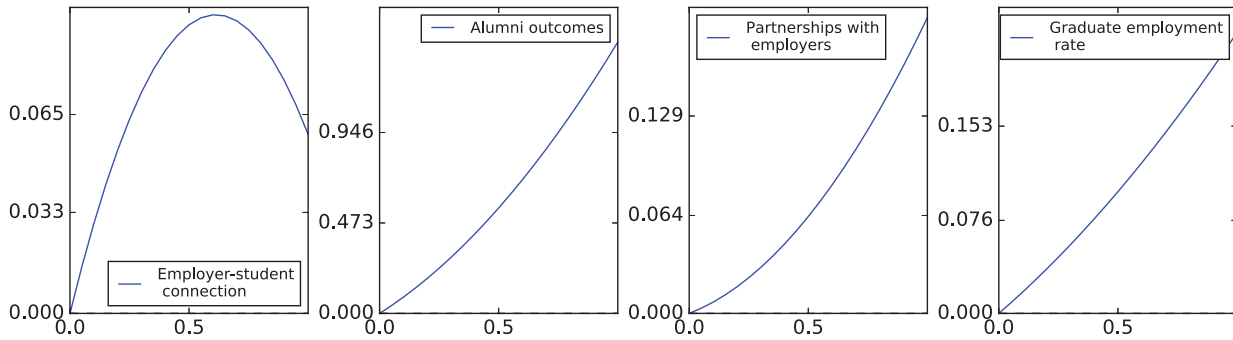


Fig. 14. Marginal value functions obtained by NN-MCDA for the university data. We use polynomial functions of degree 2 to approximate the model. The black dashed line is the baseline rate satisfying $p(\hat{y} = 1|\mathbf{x}) = p(\hat{y} = 0|\mathbf{x}) = 0.5$. Same in following Figures.

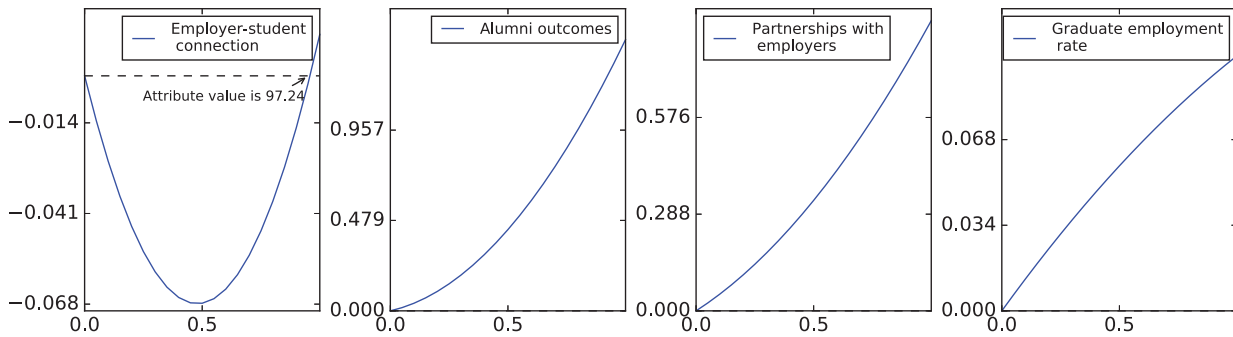


Fig. 15. Marginal value functions obtained by logistic regression model for the university data.

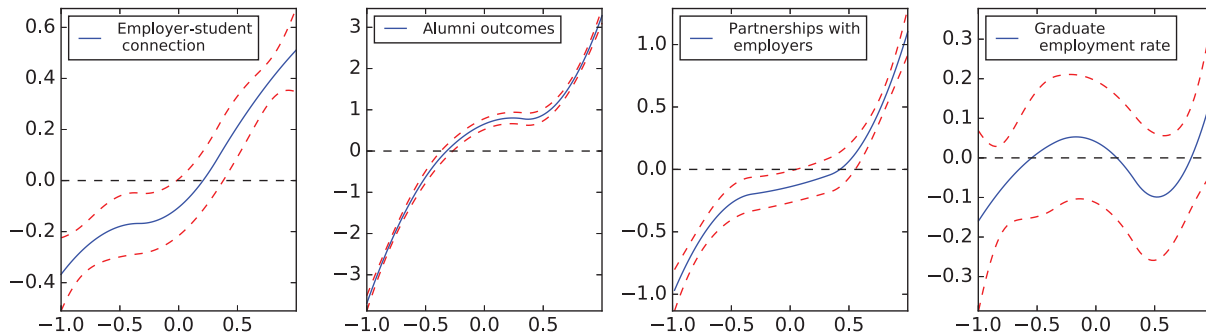


Fig. 16. Marginal value functions obtained by GAM with 5 splines that are in 2-degree polynomials for the university data. The red dashed lines are the 95% confidence bands.

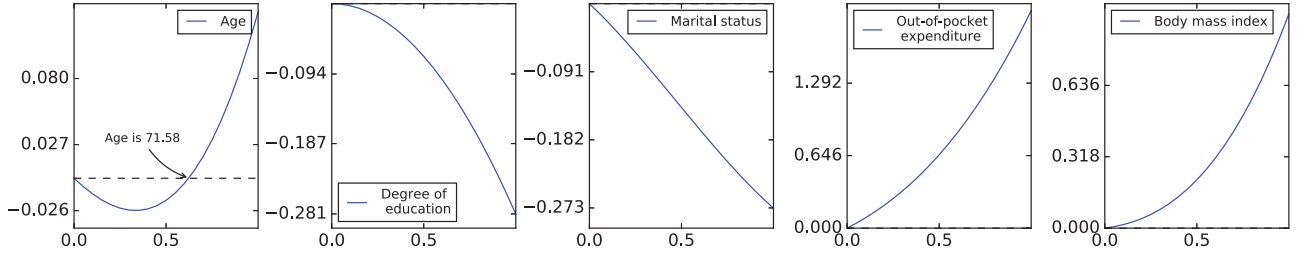


Fig. 17. Marginal value functions obtained by NN-MCDA for the depression data. From the left to right, the attributes are age, degree of education, marital status, out-of-pocket expenditure, and BMI. We use polynomial functions of degree 3 to approximate the model. The black dashed line is the baseline rate satisfying $p(\hat{y} = 1|\mathbf{x}) = p(\hat{y} = 0|\mathbf{x}) = 0.5$. Same in following figures.

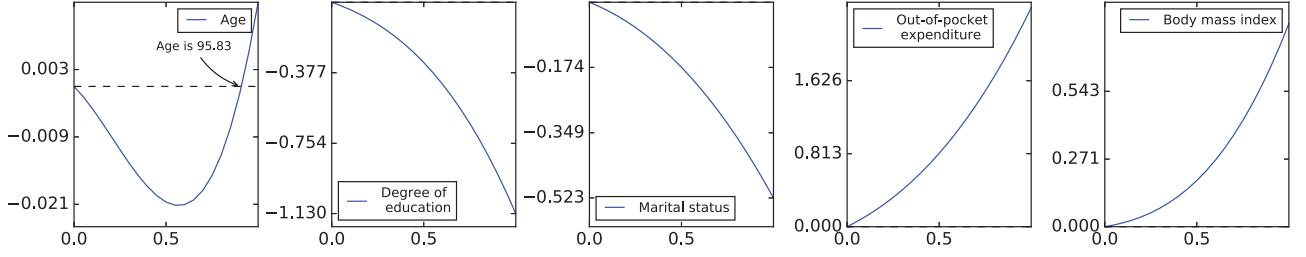


Fig. 18. Marginal value functions obtained by logistic regression model for the depression data.

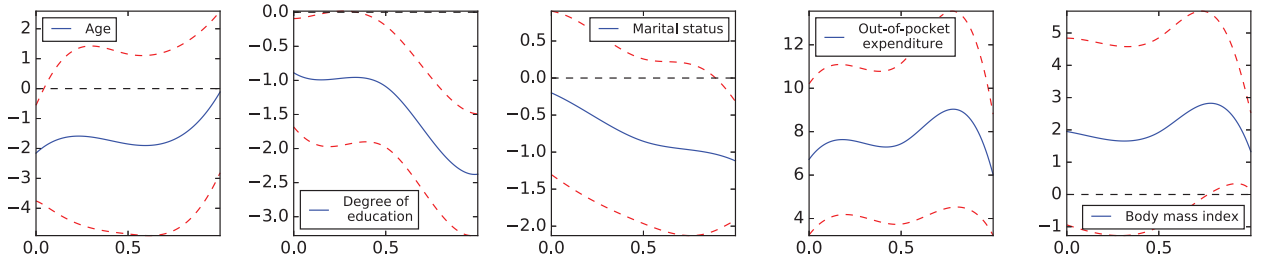


Fig. 19. Marginal value functions obtained by GAM with 5 splines that are in 3-degree polynomials for the depression data.

Table 5
Averaged AUC and α for training and testing sets.

	Training set	Testing set	α
NN-MCDA	0.678	0.669	0.393
Logistic Regression	0.621	0.593	1
MLP	0.698	0.668	0
GAM (5 splines)	0.628	0.613	–
UTADIS-5	0.601	0.579	–

The respondents with CES-D scores higher than or equal to 1 are assumed to be at risk for depression (positive samples). There are a total of 9816 positive samples and 7880 negative samples. We randomly choose 90% from both positive and negative samples to train the model. We train the NN-MCDA, MLP, GAM, logistic regression model, and UTADIS for 30 times, and present the average results in Table 5. The obtained marginal value functions are visualized in Figs. 17, 18, and 19.

We first analyze the similar conclusions by three baseline interpretable models. For the last four attributes, both NN-MCDA and logistic regression models capture similar monotonic trends. As for *degree of education* and *marital status*, the curves are under the baseline rate indicating higher education and the longer length of the marriage can reduce the risk for depression. This is consistent with the medical literature [35]. More specifically, both mod-

els find that the attributes *out-of-pocket expenditure* and *Body mass index* would increase the risk for depression. Since the obtained value functions are in a convex shape, the growth of the risk will increase along with the increase of these attribute values.

Both the NN-MCDA and logistic regression obtain a convex curve for attribute *age* with part of the curve being negative and the rest being positive (see Figs. 17 and 18). This indicates that the risk for depression does not increase while aging if the adult is younger than a threshold. The risk for depression increases fast after an adult passes an age threshold. The threshold for NN-MCDA is 71.58, which makes sense because most adults younger than 71.58 could be enjoying their retirement and their body functions do not degrade much. However, the threshold for the logistic regression is 95.83, which seems unrealistic and inconsistent with the literature [38].

Similar to the previous experiment, GAM obtains less stable curves, which are relatively more difficult to interpret. For attribute *age*, GAM obtains similar patterns as NN-MCDA and logistic regression models. For attributes *degree of education* and *marital status*, although the curves indicate that the increase in educational and marriage time results in a lower risk for depression, their shapes are less smooth than those obtained by NN-MCDA and logistic regression. As for *out-of-pocket expenditure*, GAM even obtains quite counter-intuitive results, indicating an oscillating association between the expenditure and the risk for depression.

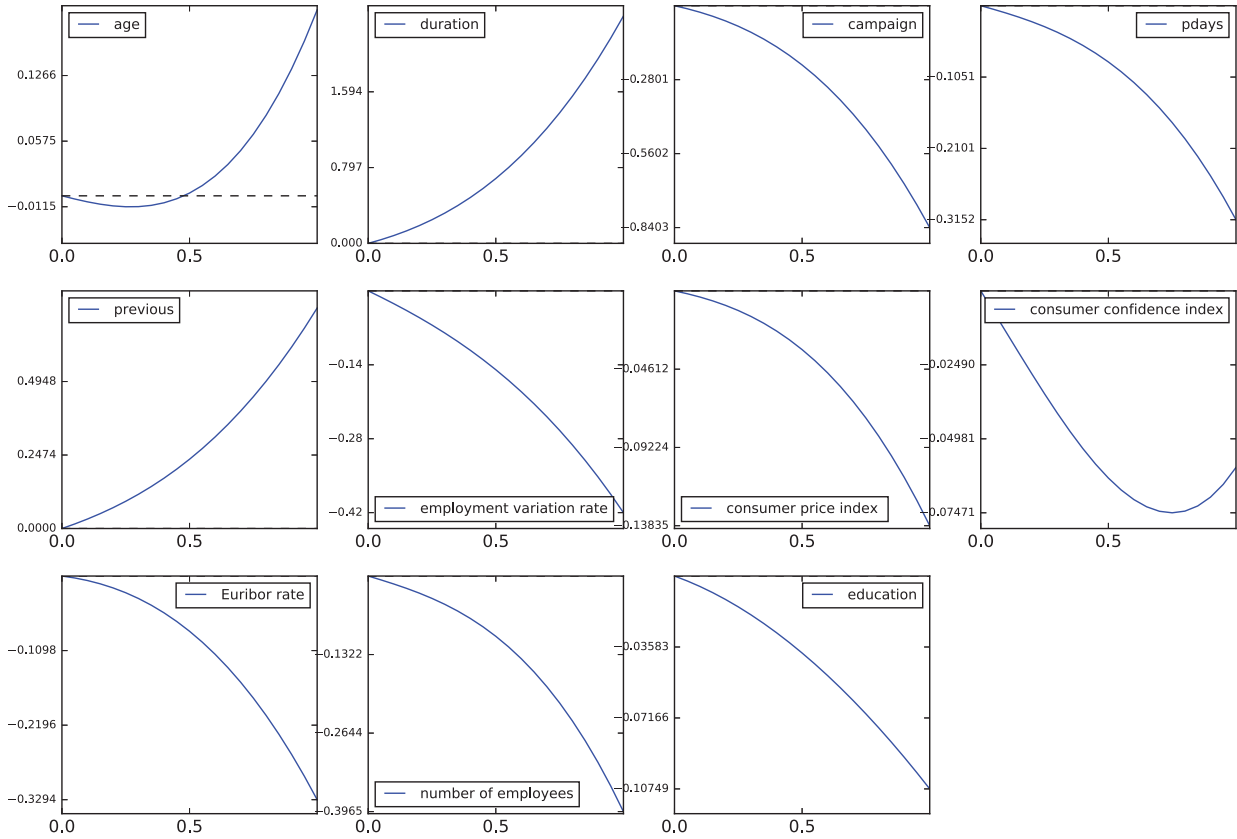


Fig. 20. Marginal value functions obtained by NN-MCDA for the telemarketing data.

3.4. Predicting the success of bank telemarketing

In this section, we employ a benchmark telemarketing dataset⁸ to further verify the efficacy of NN-MCDA. In this task, we aim to predict the success of selling bank long-term deposits by making telemarketing calls [14]. After data preprocessing, there are 40,787 records and 48 attributes (11 of them are numeric and the rest are binary). In this experiment, we focus on the contributions of 11 numeric attributes in the linear component, and import all attributes to the nonlinear component. We utilize the 3-degree polynomial functions and randomly choose 80% of data to train the NN-MCDA model for 100 times. In addition to leading to superior prediction performance (the averaged AUC is around 0.889), NN-MCDA can provide explicit marginal value functions to depict the detailed contributions of individual attributes. The performance is significantly higher than the best result 0.794 reported in [14].

Fig. 20 presents the normalized marginal value functions for all 11 numeric attributes.⁹ We find that the marginal value function of the age indicates that when a client is younger than 55, he or she is less likely to buy long-term deposits. However, over the course of aging, the client tends to save for future costs. That implies that an experienced DM (a telemarketing campaign manager) should target older clients rather than younger ones. Another interesting pattern is about the Euribor rate. NN-MCDA shows that the Euribor rate is negatively associated with the probability for long-term

deposit subscription, and such probability declines more quickly along with the increase of the Euribor rate. Consumer confidence index (CCI) measures the consumers' attitude towards the economy. The original value of the consumer confidence index ranges from -50.8 to -26.9 indicating that the clients are pessimistic towards investments, and thus they may tend to purchase less long-term deposits and keep the cash. Nevertheless, while the CCI increases, the probability of long-term deposit subscription starts to increase as well because the clients become more confident in the economy. Similarly, we can interpret the marginal value functions for all variables. Such interpretability extends the DM's understanding of the clients' investment propensity, and could inform better telemarketing strategy.

The experiments with real data demonstrate that the proposed NN-MCDA can effectively capture the patterns in human decision-making through learning the marginal value functions, which characterize the contribution of individual attributes to the predictions. The NN-MCDA presents good potential in enhancing the empirical studies through providing a detailed marginal value function instead of a single coefficient for each attribute. Moreover, the prediction performance of NN-MCDA is close to a full complexity model (MLP), and much better than that of baseline interpretable models (GAM and logistic regression model). This finding is further verified through comparing its performance with that reported in the literature.

4. Discussion

In this section, we summarize the insights from the experiments, discuss the use and extension of the proposed NN-MCDA model, and compare the proposed NN-MCDA with ensemble learning.

⁸ The dataset is available on <https://archive.ics.uci.edu/ml/datasets/Bank+Marketing>. The descriptions about the attributes and data, please refer to the website.

⁹ duration: last contact duration; campaign: number of contacts performed during this campaign and for this client; pdays: number of days that passed by after the client was last contacted from a previous campaign; previous: number of contacts performed before this campaign and for this client.

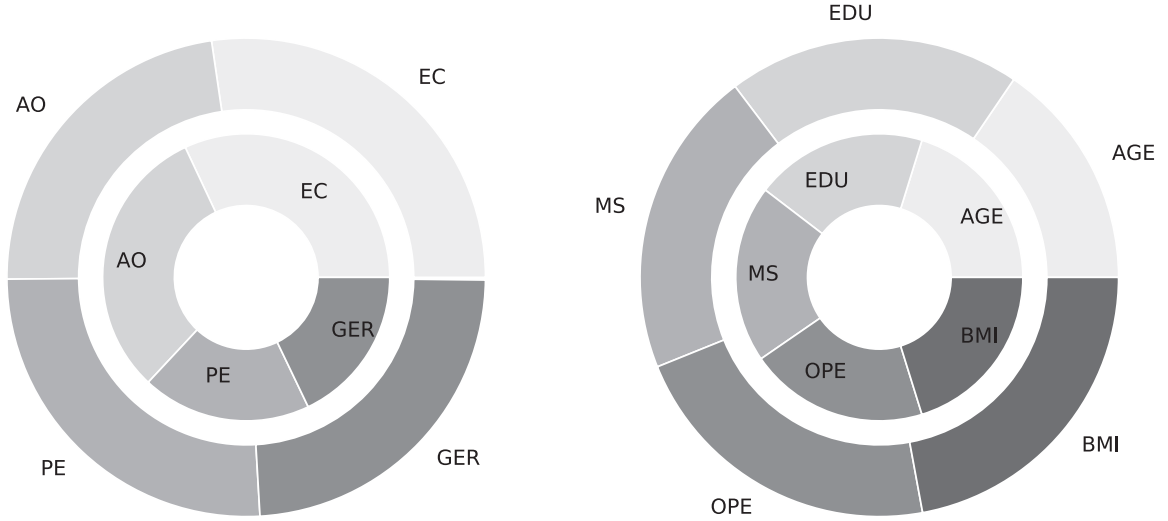


Fig. 21. Normalized attribute weights in university employee reputation ranking and geriatric depression prediction. The inner circles are normalized attribute weights that are obtained by logistic regression model and the outer ones are obtained by NN-MCDA.

4.1. Attribute importance

To explain the importance of each attribute, we present the normalized attribute weights obtained from previous experiments in Fig. 21. In the university ranking problem, both models determine that *employer-student connection* is the most important attribute, however, they give different orders of other three attributes. Given the limited resources and the obtained importance of attributes, maintaining a good employer-student connection with frequent employer presences on campus is the most effective method to achieve good employer reputation of a university.

As for the depression prediction problem, a regression model assigns almost equal importance to the five attributes, which is not intuitive for the DM. The NN-MCDA provides an order of the attributes according to the importance: $BMI \sim OPE > MS \sim EDU > AGE$. Such order suggests that obesity and loads of expenditure are the most important risk factors of becoming depressed. While estimating an old adult's risk for depression, a DM (general physician, specialists, and geriatricians) should prioritize the problems related to the older adult's body weight and economical conditions.

4.2. Interpreting the trade-off coefficient

As a key coefficient, determining the value of α is important to practical applications. It reflects the influence of high-order interactions and the complex nonlinearity of variables on the final decisions. In this regard, the NN-MCDA model can be used to explore the complexity of the learning problem. If the convergence α is very small, it indicates that the data is highly complex and the assumption of preference independence is not valid. The DM should then use the latest models that account for attribute interactions, such as a Choquet integral-based model [39] or the full complexity machine learning models. On the contrary, if α is close to 1, the DM is recommended to use a simpler model to avoid massive computational time and non-interpretable results.

In the case of prediction for depression (Table 5), the convergence α is around 0.4. It indicates that the involved attributes are possibly interacting in this problem. Some related medical studies also empirically demonstrated some interactions between attributes, for example, older adults that are relatively young with a higher degree of education may be involved in more social activities and have a more contented life, which leads to lower risk for depression [40]. Moreover, these older adults with the longer mar-

riage will obtain more family support and thus they have a lower risk for depression [41]. Given convergence α and marginal value functions obtained in the presence of high-order correlations among attributes, we could develop algorithms to use lower-ordered attribute interactions, e.g.: pairwise and triple-wise interactions, to approximate the higher ones. The framework can then be extended to a two-step procedure, including determining marginal value functions and deducing possible lower-ordered correlations.

4.3. Extending the NN-MCDA framework

The proposed NN-MCDA presents a general modeling framework, which can be easily extended to enhance the performance and adaptivity for various problems. In this section, we discuss the two extensions, including replacing the model in the nonlinear component and incorporating attributes in the nonlinear component.

4.3.1. Replace the model in the nonlinear component

Given different types of datasets, the proposed NN-MCDA model can be modified by replacing the neural networks in the nonlinear component by other network structures. In Section 3.1.1, NN-MCDA has difficulty in handling extremely complex data ($\mathcal{D}_{polynomial-3}^n$ and $\mathcal{D}_{polynomial-15}^n$). To improve the performance of

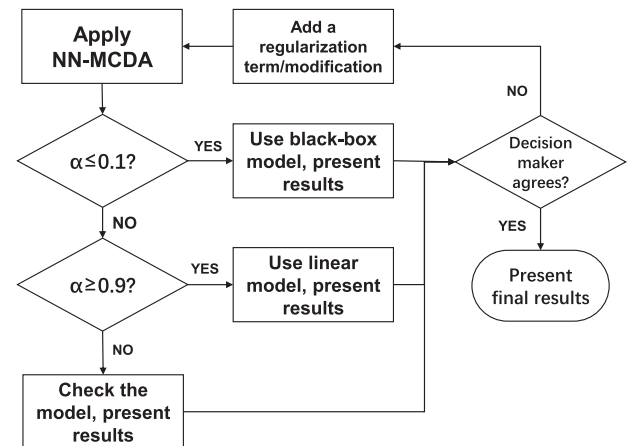


Fig. 22. A heuristic user-interactive process for modifying the proposed NN-MCDA.

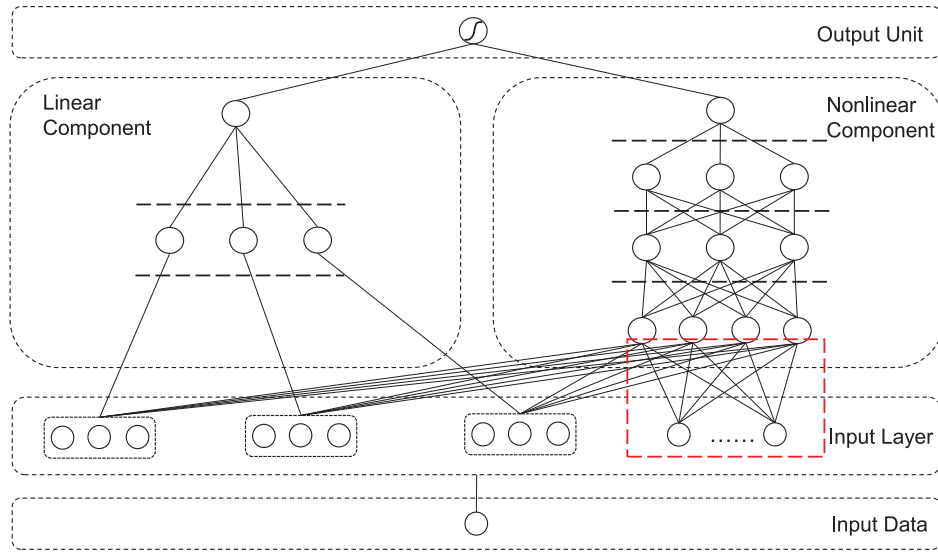


Fig. 23. A new framework for NN-MCDA considering more attributes in the nonlinear component.

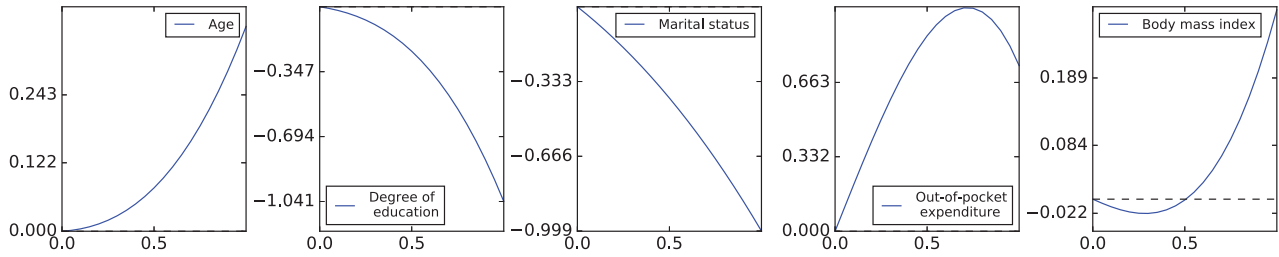


Fig. 24. For depression data, the marginal value function obtained by linear component while adding two more attributes to the nonlinear component.

this data, we can introduce more layers in the MLP or increase the number of neurons in each layer. For an image classification problem, we can replace the MLP with a convolutional neural network (CNN), and use the features obtained from CNN as the input for the linear component. To fit time-series or free text data, we can replace the MLP with a recurrent neural network.

In addition, the proposed model can be progressively modified by iteratively interacting with the DM. We provide a user-interactive process to determine the ultimate model. The framework is shown in Fig. 22 and explained as follows:

- Step 1. We first apply the NN-MCDA model to the management problem. While the model converges, we obtain the value of α .
- Step 2. If $\alpha > 0.1$, go to step 3. If $\alpha \leq 0.1$, which indicates that the data are potentially very complex. We opt for a full complexity black-box model to achieve higher accuracy and present the results to the DM. If the DM agrees to use the black-box model, the process ends. Otherwise, we add a regularization term (e.g.: use $MSE_1 = MSE + (1 - \alpha)^2$)¹⁰ to the original NN-MCDA, and go back to Step 1.
- Step 3. If $\alpha < 0.9$, go to step 4. If $\alpha \geq 0.9$, which indicates that a simple additive model is sufficient to fit the data, we explain the results to the DM. If the DM is satisfied with the accuracy, the process ends. Otherwise, we can modify

the NN-MCDA model by increasing the complexity of the nonlinear component (e.g.: using deeper MLP or other neural net-based models). Then, we go back to Step 1.

- Step 4. If $0.1 < \alpha < 0.9$, we present the underlying model and results to the DM. If there are no further requirements, the process ends. If the DM requires further modifications (such as adding regularization terms or modifying the nonlinear component), we modify the model accordingly and then go to Step 1.

4.3.2. Flexible inclusion of attributes in the nonlinear component

In practice, human decision-making usually focuses on a small number of well-chosen attributes/criteria [3]. However, there could exist other minor attributes that do not directly contribute to the prediction but could affect the prediction through non-traceable complex interactions with other attributes (for example, the interaction between the nonlinear transformation of an attribute and the nonlinear transformation of five other attributes). These minor attributes can be incorporated by the nonlinear component.

In the geriatric depression experiment, the gender of an older adult may not directly indicate a difference in the risk for depression, however, it might still influence the prediction through complex interactions with other attributes. We further extend the NN-MCDA model to incorporate gender and smoking status into the nonlinear component (as shown in Fig. 23). We find that the incorporation of these attributes indeed improves the prediction accuracy (the AUC for testing set increases from 0.669 to 0.675), while still maintains the similar marginal value functions in the linear component (see Fig. 24). If we add two more attributes, for instance, whether the respondent received any home cares in

¹⁰ The added regularization term $(1 - \alpha)^2$ allocates more weight to the linear component at the cost of lower fitting accuracy. We can also change the regularization term to $(2\alpha - 1)^2$, which leads to a balanced model that favors a model with equal weights to the linear and nonlinear components. The exact form of the loss function should be selected according to the problem settings.

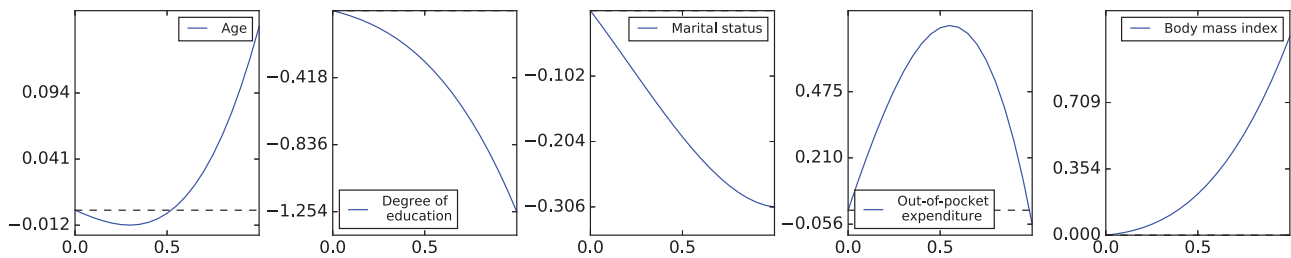


Fig. 25. For depression data, the marginal value function obtained by linear component while adding four more attributes to the nonlinear component.

last two years and whether the health problem limited his/her work, the AUC increases more obviously (from 0.675 to 0.708) and the marginal value functions still provide convincing results (see Fig. 25).

Note that, the curves for *OPE* in Figs. 24 and 25 are still positive, indicating that *OPE* is consistently positively associated with the risk for depression (similar to Fig. 17). Interestingly, by introducing additional binary attributes, the curve for *OPE* changes from a monotonically increasing shape (see Fig. 17) to a positive inverted-U shape (see Figs. 24 and 25). This indicates that *OPE* could have interactions with the newly added binary attributes, which affects the main effect of *OPE* in Fig. 17.

To explore attribute interactions, we can analyze the parameters in the nonlinear part after the model is learned. In Appendix E, we incorporate a state-of-the-art method to identify the attribute interactions in the proposed NN-MCDA and present the most important pairwise interactions in the geriatric depression experiment.

4.4. Joint training process

In the NN-MCDA model, the linear and nonlinear components are combined by a trade-off coefficient α . Their sum is then fed to a common logistic function for a joint training process. Note that this joint training process is different from ensemble learning [42], in which multiple classifiers are trained individually and their predictions are simply combined after every model is optimized separately. For example, an ensemble learning approach could have a linear logistic regression model and an MLP model to make predictions for the same dataset separately, and then integrate the prediction results of the two models. The joint training process indicates that the linear and nonlinear components are connected. While we tune the parameters in one component, the other component will be affected. If the model is at its global optimal, the predictions can be made.

5. Conclusion and future work

In this paper, we proposed a framework for a novel hybrid machine learning model, namely the NN-MCDA, which combines the traditional MCDA model and neural networks. MCDA uses marginal value functions to describe the explicit contribution of individual attributes to the predictions, while the neural network considers the implicit high-order interrelationships among attributes. The framework automatically balances the trade-off between two components. NN-MCDA is more interpretable than a full complexity model and maintains similar predictability.

We present simulation experiments to demonstrate the effectiveness of NN-MCDA. The experiments show that (1) polynomial of higher degrees does not always improve on accuracy; (2) There is a trade-off between the interpretability and the predictability of the model. NN-MCDA can achieve a good balance between them; (3) Given simple data, NN-MCDA performs as good as an interpretable model, while given more complex data, NN-MCDA outperforms an interpretable model. We also present how to apply

the NN-MCDA framework to real-world decision-making problems. These experiments with real data demonstrate the good prediction performance of NN-MCDA and its ability in capturing the detailed contributions of individual attributes.

We envisage the following directions for future researches based on the NN-MCDA framework. First, we can further enhance the interpretability of the model through proposing algorithms to approximate the attribute interactions after obtaining the marginal value functions. Second, further studies are needed to validate the effectiveness of the NN-MCDA variants that were introduced in the discussion section. Third, we plan to explore how to obtain stable marginal value functions using a more theoretical approach. A possible way is to learn from the robust ordinal regression models [43]. Last, but not the least, applying heuristic algorithms, for instance, a genetic algorithm to automatically determine the best degree of polynomials constitutes another interesting direction for future work.

CRedit authorship contribution statement

Mengzhuo Guo: Conceptualization, Methodology, Investigation, Writing - original draft, Writing - review & editing, Visualization. **Qingpeng Zhang:** Conceptualization, Methodology, Investigation, Writing - original draft, Writing - review & editing, Supervision, Funding acquisition. **Xiuwu Liao:** Methodology, Investigation, Writing - review & editing, Supervision, Funding acquisition. **Frank Youhua Chen:** Investigation, Writing - review & editing. **Daniel Dajun Zeng:** Investigation, Writing - review & editing.

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Appendix A. Proof of Proposition 1

Appendix B. Proof of Proposition 2

Appendix C. Computation of AUC

Appendix D. Full simulation results

Appendix E. A method to detect pairwise interactions

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.omega.2020.102263](https://doi.org/10.1016/j.omega.2020.102263).

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